

Past Problems

for Korean Undergraduate Mathematics Competition

Jungwoo Kim

Department of Mathematics

Pohang University of Science and Technology

This document collects past problems from the Korean Undergraduate Mathematics Competition. General mathematical notation and writing follow *(MATH000) Mathematical Language*.

2001.2.1. Let $0 < a, b < 1$ satisfy

$$\int_0^a \frac{1}{\sqrt{1-x^2}} dx = \int_b^1 \frac{1}{\sqrt{1-x^2}} dx.$$

Find $a^2 + b^2$.

2001.2.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

2001.2.3. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2y = 0, \quad x > 0.$$

Hint: $y = e^{x^2}$ is a solution of the differential equation.

2001.2.4. Let $f: [0, \infty) \rightarrow [0, \infty)$ satisfy

$$2f(x)f(1) \leq f(x)f(1) + 1 \leq 2f(2x)$$

for every $x \geq 0$. Prove that f is constant.

2001.2.5. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative. Evaluate

$$\lim_{\lambda \rightarrow \infty} \int_a^b f(x) \left(|\sin(\lambda x)| - \frac{2}{\pi} \right) dx.$$

2001.2.6. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C , respectively. Find the maximum value of

$$a \cos A + b \cos B + c \cos C.$$

2001.2.7. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.2.8. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ be a unit vector. Prove that

$$\sum_{i=1}^n \|\mathbf{x} - \mathbf{v}_i\|^2 \geq 2n - 2\sqrt{n}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

2001.2.9. Let $\lambda \geq 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2001.1.1. Show that there are infinitely many triples $(a, b, c) \in \mathbb{N}^3$ satisfying

$$a \cos\left(\arctan \frac{b}{a}\right) + b \sin\left(\arctan \frac{b}{a}\right) = c.$$

2001.1.2. For $n \times n$ matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$, prove that

$$\det \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{I}_n \end{pmatrix} = \det(\mathbf{A} - \mathbf{BC}),$$

where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

2001.1.3. Find the general solution of

$$y'' - \frac{1}{x}y' - 4x^2y = 0, \quad x > 0.$$

Hint: $y = e^{x^2}$ is a solution of the differential equation.

2001.1.4. Let $f: [0, \infty) \rightarrow [0, \infty)$ satisfy

$$2f(x)f(1) \leq f(x)f(1) + 1 \leq 2f(2x)$$

for every $x \geq 0$. Prove that f is constant.

2001.1.5. Evaluate

$$\lim_{n \rightarrow \infty} \int_0^{\pi/2} \frac{\sin x}{n^2 \left(x - \frac{\pi}{4}\right)^2 + 1} dx.$$

2001.1.6. A triangle ABC is inscribed in a circle of radius 1. Let a, b, c be the side lengths opposite the angles A, B, C , respectively. Find the maximum value of

$$a \cos A + b \cos B + c \cos C.$$

2001.1.7. Evaluate the infinite series

$$\sum_{k=0}^{\infty} (-1)^k \left(\frac{1}{2}\right)^{3k+2} \frac{9k+5}{(3k+1)(3k+2)}.$$

2001.1.8. Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \in \mathbb{R}^2$ satisfy

$$\|\mathbf{v}_i\| \leq 1 \quad (i = 1, 2, \dots, n).$$

Prove that there always exist signs

$$\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n \in \{1, -1\}$$

such that

$$\|\varepsilon_1 \mathbf{v}_1 + \varepsilon_2 \mathbf{v}_2 + \dots + \varepsilon_n \mathbf{v}_n\| \leq \sqrt{2}.$$

Here $\|\mathbf{v}\|$ denotes the Euclidean norm of $\mathbf{v}$.

2001.1.9. Let $\lambda \geq 0$ and $0 < p < 1$. Suppose f is defined for $x > 0$ and satisfies

$$f(x) = (x - \lambda f(x))^p.$$

Let $g(x) = x^p$. Prove that, for every $x > 0$,

$$\frac{f'(x)}{g'(x)} < \frac{1}{p}.$$

2002.0.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Suppose the function $x \mapsto xf(x)$ is differentiable for $x \neq 0$ and

$$\frac{d}{dx}(xf(x)) = e^x.$$

Find f .

2002.0.2. Let (a_n) be an arithmetic sequence with first term a and common difference d , and define the 2002×2002 matrix

$$\mathbf{A} := \begin{pmatrix} a_1 & a_2 & a_3 & a_4 & \cdots & a_{2002} \\ a_2 & a_1 & a_2 & a_3 & \cdots & a_{2001} \\ a_3 & a_2 & a_1 & a_2 & \cdots & a_{2000} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{2002} & a_{2001} & a_{2000} & a_{1999} & \cdots & a_1 \end{pmatrix}.$$

Express $\det(\mathbf{A})$ in terms of a and d .

2002.0.3. Let $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ be unit vectors in $\mathbb{R}^n$. Prove that

$$|\langle \mathbf{a}, \mathbf{c} \rangle \langle \mathbf{b}, \mathbf{d} \rangle - \langle \mathbf{a}, \mathbf{d} \rangle \langle \mathbf{b}, \mathbf{c} \rangle| \leq 1,$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^n$.

2002.0.4. Let $n \in \mathbb{N}$, and let $a_1, \dots, a_n, x_1, \dots, x_n$ be positive real numbers satisfying

- (i) $a_1 \leq a_2 \leq \dots \leq a_n$;
- (ii) $x_k \geq x_n$ for $k = 1, 2, \dots, n-1$.

Prove that

$$(a_1 + x_1 a_2)(a_2 + x_2 a_3) \cdots (a_{n-1} + x_{n-1} a_n)(a_n + x_n a_1) \geq a_1 a_2 \cdots a_n (x_1 + 1)(x_2 + 1) \cdots (x_n + 1).$$

2002.0.5. For $\alpha \in \mathbb{C}$, define the $n \times n$ matrix $\mathbf{A}_n = (a_{ij})$ by

$$a_{ij} = \begin{cases} 1, & i = j, \\ \alpha, & i - j = -1, \\ \bar{\alpha}, & i - j = 1, \\ 0, & |i - j| \geq 2. \end{cases}$$

Prove that $\mathbf{A}_n$ is positive semidefinite for every $n = 1, 2, \dots$ if and only if

$$|\alpha| \leq \frac{1}{2}.$$

Here $\bar{\alpha}$ denotes the complex conjugate of α .

2002.0.6. Solve the system of differential equations

$$x''(t) + 2y'(t) - x(t) = 0, \quad y''(t) - 2x'(t) - y(t) = 0$$

subject to

$$x(0) = y(0) = 0, \quad x'(0) = a > 0, \quad y'(0) = 0.$$

2002.2.1. For $(x, y) \neq (0, 0)$, define

$$f(x, y) = \frac{x^2 - xy + y^2}{2x^2 + y^2}.$$

Find the minimum value of f .

2002.2.2. Let $f, g, h: [-1, 1] \rightarrow \mathbb{R}$ be continuous. Suppose that f is odd, g is even, and

$$\int_{-1}^1 f(x)^2 \, dx = \int_{-1}^1 g(x)^2 \, dx = 1.$$

Prove that

$$\left(\int_{-1}^1 f(x)h(x) \, dx \right)^2 + \left(\int_{-1}^1 g(x)h(x) \, dx \right)^2 \leq \int_{-1}^1 h(x)^2 \, dx.$$

2002.2.3. Let $a > b > 0$. Evaluate

$$\int_0^\infty \frac{\arctan(ax) - \arctan(bx)}{x} dx.$$

2002.1.1. Let $f: [a, b] \rightarrow \mathbb{R}$ have a continuous derivative, with $f(a) = 0$. Suppose that

$$\left| \frac{f'(x)}{2003} \right| \leq \left| \frac{f(x)}{2002} \right| \quad (x \in [a, b]).$$

Prove that $f(x) \equiv 0$ on $[a, b]$.

2002.1.2. Let $n \in \mathbb{N}$ with $n > 1$. Prove that

$$\frac{((n-1)!)^{1/(n-1)}}{(n!)^{1/n}} < \frac{(n!)^{1/n}}{((n+1)!)^{1/(n+1)}}.$$

It is given that

$$4 > e^{4-2\sqrt{2}}.$$

2002.1.3. Let

$$\alpha = 0.0110101000101000101 \dots,$$

where the n th digit after the decimal point is 1 if n is prime and 0 otherwise. Prove that α is irrational.

2003.2.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2003.2.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC , BCD , CDA , and DAB , respectively. Prove that $PQRS$ is a rectangle.

2003.2.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.2.4. Let (x_n) be a sequence satisfying

$$0 \leq x_{n+1} \leq x_n$$

for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} x_n$$

converges. Determine

$$\lim_{n \rightarrow \infty} nx_n.$$

2003.2.5. Let $\mathbf{x} = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ satisfy

$$x_1 + x_2 + \dots + x_n = 0.$$

Prove that

$$\max_{1 \leq j \leq n} |x_j| \leq \left(\frac{n-1}{n} \right)^{1/2} \|\mathbf{x}\|_2,$$

where

$$\|\mathbf{x}\|_2 = (x_1^2 + x_2^2 + \dots + x_n^2)^{1/2}.$$

2003.2.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k - 1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.2.7. Prove that the real number $\sin 1$ is irrational.

2003.2.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3} \right)^n (n+1)$$

for every $x \in A$.

2003.2.9. Let f be the 2π -periodic function defined by

$$f(x) = \begin{cases} \pi + x, & -\pi < x \leq 0, \\ \pi - x, & 0 < x \leq \pi. \end{cases}$$

Using the Fourier series of f , prove that

$$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \dots + \frac{1}{(2n-1)^2} + \dots.$$

2003.1.1. Find the value of the natural number

$$\sum_{k=1}^{\infty} \binom{10^{10}}{\lfloor \log_{11} k \rfloor}.$$

Here $\binom{n}{r}$ denotes the binomial coefficient and $\lfloor x \rfloor$ denotes the greatest integer not exceeding x .

2003.1.2. Four points A, B, C, D are given in clockwise order on a fixed circle, and the quadrilateral $ABCD$ contains the center of the circle. Let P, Q, R, S be the incenters of the triangles ABC, BCD, CDA , and DAB , respectively. Prove that $PQRS$ is a rectangle.

2003.1.3. Let $a, b \in \mathbb{N}$ and let p, q be primes with $p < 2003$ and $q < 2003$. Find all ordered pairs

$$(p^a, q^b)$$

satisfying

$$p^a - q^b = 1.$$

2003.1.4. Let (a_n) be a sequence with $a_n \geq 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n$$

converges. Prove that, for every $p > \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

converges. Also give an example showing that, when $p = \frac{1}{2}$, the series

$$\sum_{n=1}^{\infty} \frac{\sqrt{a_n}}{n^p}$$

may diverge.

2003.1.5. Let $p \geq 1$ be a real number, and let $x_1, x_2, \dots, x_n > 0$ satisfy

$$x_1 + x_2 + \dots + x_n = 1.$$

Prove that

$$\sum_{j=1}^n \left(x_j + \frac{1}{x_j} \right)^p \geq \frac{(1+n^2)^p}{n^{p-1}}.$$

2003.1.6. For each $k = 1, 2, \dots, 2002$, let x_k be a real number satisfying

$$k-1 \leq x_k \leq k.$$

Define

$$B_k = \left\{ \frac{2}{k} \sum_{i=1}^k |x - x_i| \mid x \in \mathbb{R} \right\}$$

and

$$A = \left\{ \sum_{k=1}^{2002} y_k \mid y_k \in B_k, k = 1, 2, \dots, 2002 \right\}.$$

Prove that

$$\binom{2003}{2} \in A.$$

2003.1.7. Prove that the real number $\sin 1$ is irrational.

2003.1.8. Let $f: [0, 3] \rightarrow \mathbb{R}$ be continuous and suppose that

$$\int_0^3 x^k f(x) \, dx = 0 \quad (k = 0, 1, 2, \dots, n-1)$$

and

$$\int_0^3 x^n f(x) \, dx = 3.$$

Prove that there exists an interval $A \subseteq [0, 3]$ such that

$$|f(x)| \geq \left(\frac{2}{3}\right)^n (n+1)$$

for every $x \in A$.

2003.1.9. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix whose entries are all 0 or 1. Suppose that, for some $t \in \mathbb{N}$,

$$\mathbf{A}\mathbf{A}^T = \begin{pmatrix} t+1 & 1 & \cdots & 1 & 1 \\ 1 & t+1 & \cdots & 1 & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 1 & 1 & \cdots & t+1 & 1 \\ 1 & 1 & \cdots & 1 & t+1 \end{pmatrix} = t\mathbf{I}_n + \mathbf{J},$$

where $\mathbf{A}^T$ denotes the transpose of $\mathbf{A}$, $\mathbf{I}_n$ is the $n \times n$ identity matrix, and $\mathbf{J}$ is the $n \times n$ all-ones matrix. Express $\det(\mathbf{A})$ as a function of t .

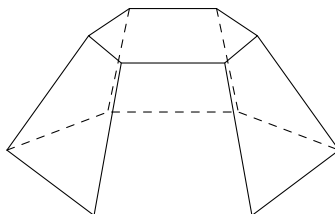
2004.2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$. Prove the following.

- (a) If f is continuous at 0, then f is continuous at every $x \in \mathbb{R}$.
- (b) If f is continuous on $\mathbb{R}$ and $f(1) = c$, then $f(x) = cx$ for every $x \in \mathbb{R}$.

2004.2.2. A frustum of a pyramid has parallel upper and lower bases and uniform density. Find the ratio of the distance from its center of mass to the lower base to the distance from its center of mass to the upper base. The similarity ratio of the upper base to the lower base is r , where $r < 1$.



2004.2.3. A convex quadrilateral $ABCD$ has side lengths a, b, c, d , in this order starting from AB . Show that if its area is maximal among all such quadrilaterals, then $ABCD$ is cyclic.

2004.2.4. Let $(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ be an orthonormal basis of $\mathbb{R}^n$. Find the orthogonal complement of

$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.2.5. Let $f: [a, c] \rightarrow (0, \infty)$ be continuous on $[a, c]$ and differentiable on (a, c) . Let

$$P = (x_0, f(x_0))$$

be a point on the graph of f . Let α and γ be the angles made with the x -axis by the lines joining P to $(a, 0)$ and $(c, 0)$, respectively, and let β be the angle made with the x -axis by the tangent line to the graph at P . Show that there exists $x_0 \in (a, c)$ such that

$$\tan \alpha + \tan \beta + \tan \gamma = 0.$$

If the tangent line does not meet the x -axis, take $\beta = 0$.

2004.2.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

$$\sum_{n=1}^{\infty} a_n$$

diverges, and the series

$$\sum_{n=1}^{\infty} a_n^2$$

converges. If

$$\sum_{n=1}^{\infty} f(a_n)$$

converges, prove that

$$\sum_{n=1}^{\infty} |f(a_n)|$$

also converges.

2004.2.7. Suppose a plane curve $(x(t), y(t))$ of period T satisfies

$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy, \quad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.2.8. Evaluate

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx.$$

2004.2.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

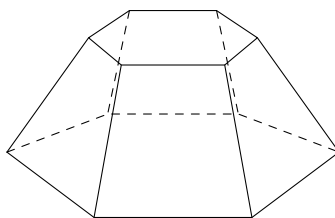
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$$\text{span}\{i\mathbf{v}_i - j\mathbf{v}_j \mid 1 \leq i < j \leq n\}.$$

2004.1.5. Let A be the least positive integer of the form

$$33^k - 7^\ell, \quad k, \ell \in \mathbb{N},$$

and let B be the least positive integer of the form

$$7^m - 33^n, \quad m, n \in \mathbb{N}.$$

Find $A + B$.

2004.1.6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function whose second derivative f'' exists and is bounded on $[0, 1]$. Let (a_n) be a sequence such that $0 < a_n < 1$ for every $n \in \mathbb{N}$, the series

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diverges, and the series

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$$\frac{dx}{dt} = ax - bxy, \quad \frac{dy}{dt} = -cy + dxy, \quad x(0) = y(0) = 1,$$

where $a, b, c, d > 0$. Find

$$\bar{x} := \frac{1}{T} \int_0^T x(t) \, dt, \quad \bar{y} := \frac{1}{T} \int_0^T y(t) \, dt.$$

2004.1.8. Let $\mathbf{v} \in \mathbb{R}^n$ be a unit vector, and let $\mathbf{P}_{\mathbf{v}}$ denote the orthogonal projection onto the line $\text{span}(\mathbf{v})$, so that

$$\mathbf{P}_{\mathbf{v}}\mathbf{x} = (\mathbf{x} \cdot \mathbf{v})\mathbf{v}$$

for every $\mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{A}$ be an $n \times n$ positive definite symmetric matrix. Find the maximum value of λ such that

$$\mathbf{A} - \lambda \mathbf{P}_{\mathbf{v}} \succeq 0.$$

Here $\mathbf{B} \succ 0$ means that every eigenvalue of the symmetric matrix $\mathbf{B}$ is positive, and $\mathbf{B} \succeq 0$ means that every eigenvalue is nonnegative.

2004.1.9. Let B be any 30-element subset of $\{1, 2, \dots, 39, 40\}$. Prove that there exists a subset $A \subseteq B$ such that

- (i) $|A| \geq 10$;
- (ii) $x + y \notin A$ for all $x, y \in A$.

2005.0.1. Let

$$\mathbf{A} = \frac{1}{2} \begin{pmatrix} 0 & \sqrt{2} \\ -\sqrt{2} & 0 \end{pmatrix}.$$

Find the matrix

$$e^{\mathbf{A}} = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!}.$$

2005.0.2. Solve the differential equation

$$y'(t) + 2ty(t) = ty(t)^2, \quad y(1) = 1.$$

2005.0.3. Let $p \geq 3$ be an arbitrary prime, and let

$$\mathbb{F}_p = \{0, 1, \dots, p-1\}.$$

Show that the set

$$S = \{(x, y) \in \mathbb{F}_p^2 \mid x^2 + y^2 \equiv -1 \pmod{p}\}$$

is nonempty.

2005.0.4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that

$$|f(n)| \leq \frac{1}{(1+n)^{5/4}}$$

for every nonnegative integer n , and that

$$|f'(x)| \leq e^{-x}$$

for every $x \in \mathbb{R}$. Prove that

$$\int_0^{\infty} |f(x)| \, dx < \infty.$$

2005.0.5. Let S_n be the set of all bijections from $\{1, 2, \dots, n\}$ to itself. For $\sigma \in S_n$, let $f(\sigma)$ denote the number of fixed points of σ , that is,

$$f(\sigma) = |\{x \in \{1, 2, \dots, n\} \mid \sigma(x) = x\}|.$$

Find

$$\sum_{\sigma \in S_n} f(\sigma).$$

2005.2.1. A population consists of smokers and nonsmokers. Each year, 5% of the nonsmokers begin smoking and 10% of the smokers quit smoking. Prove that, after a sufficiently long time, the ratio of smokers to nonsmokers converges to a constant, and find that ratio.

2005.2.2. Let X be the set of all invertible 2×2 matrices and Y the set of all invertible 3×3 matrices. Define $\Phi: X \rightarrow Y$ by

$$\Phi \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right) = \begin{pmatrix} a^2 & 2ab & b^2 \\ ac & ad + bc & bd \\ c^2 & 2cd & d^2 \end{pmatrix}.$$

Suppose $\mathbf{A} \in X$ has eigenvalues $1, x$ and $\mathbf{B} \in X$ has eigenvalues $1 + x, 1 - x$. Find

$$\det(\Phi(\mathbf{AB})).$$

2005.2.3. Words of length n are formed using the three letters A, B, C , with the restriction that A may not occur twice consecutively. Express the number of such words as a function of n .

2005.2.4. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be twice differentiable and suppose that

$$|f(x)| \leq 1, \quad |f''(x)| \leq 1$$

for every $x \in (0, \infty)$. Prove that

$$|f'(x)| \leq 2$$

for every $x \in (0, \infty)$.

2005.1.1. A population consists of smokers and nonsmokers. Each year, 5% of the nonsmokers begin smoking and 10% of the smokers quit smoking. Prove that, after a sufficiently long time, the ratio of smokers to nonsmokers converges to a constant, and find that ratio.

2005.1.2. Let $f: [0, 1] \rightarrow \mathbb{R}$ be differentiable, with $f(0) = 0$ and $f(1) = 1$. Prove that, for every $n \in \mathbb{N}$, there exist distinct points $x_1, x_2, \dots, x_n \in [0, 1]$ such that

$$\sum_{k=1}^n \frac{1}{f'(x_k)} = n.$$

2005.1.3. Words of length n are formed using the three letters A, B, C , with the restriction that A may not occur twice consecutively. Express the number of such words as a function of n .

2005.1.4. Let (a_n) and (b_n) be sequences satisfying $a_n > 0$ and $b_n > 0$ for every $n \in \mathbb{N}$, and suppose that

$$\sum_{n=1}^{\infty} a_n = 3, \quad \sum_{n=1}^{\infty} b_n = 4.$$

Find the supremum and infimum of all possible values of

$$\sum_{n=1}^{\infty} \frac{a_n^2}{a_n + b_n}.$$

2006.0.1. Let $f(x) = \arctan(2x)$. Find $f^{(99)}(0)$.

2006.0.2. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous periodic function with period $2T$. Prove that there exists a suitable $x_0 \in \mathbb{R}$ such that

$$f(x_0 + T) = f(x_0).$$

2006.0.3. Let $\mathbf{A}$ be an $n \times n$ matrix such that

$$\text{tr}(\mathbf{X}\mathbf{Y}\mathbf{A}) = \text{tr}(\mathbf{Y}\mathbf{X}\mathbf{A})$$

for every pair of $n \times n$ matrices $\mathbf{X}, \mathbf{Y}$. Prove that $\mathbf{A}$ is a scalar multiple of the identity matrix. Here tr denotes the sum of the diagonal entries.

2006.0.4. Let (a_n) be a sequence of positive real numbers, and suppose that the limit

$$\lim_{n \rightarrow \infty} n \left(\frac{a_n}{a_{n+1}} - 1 \right)$$

exists. Prove that if this limit is greater than 1, then the series

$$\sum_{n=1}^{\infty} a_n$$

converges.

2006.0.5. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be differentiable, suppose that f' is continuous, and assume that

$$\lim_{x \rightarrow \infty} (f(x) + f'(x)) = 0.$$

Prove that

$$\lim_{x \rightarrow \infty} f(x) = 0.$$

2006.2.1. Evaluate

$$\int_{-\pi}^{\pi} \ln \left| \sin \frac{\theta}{2} \right| d\theta.$$

2006.2.2. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$ be a unit vector, and let $\mathbf{X} = (x_{ij})$ be the $n \times n$ matrix defined by

$$x_{ij} = \delta_{ij} + a_i a_j,$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Compute $\det(\mathbf{X})$.

2006.2.3. Let $h: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\int_0^{\infty} h(t) dt = 1.$$

Evaluate

$$\iint_{\mathbb{R}^2} h(x^2 + 2xy + 5y^2) dA.$$

2006.2.4. Evaluate the series

$$1 - \frac{1}{5} + \frac{1}{7} - \frac{1}{11} + \cdots + \frac{1}{6n-5} - \frac{1}{6n-1} + \cdots.$$

2006.1.1. Prove that the improper integral

$$\int_0^\infty \sin(x^2) \, dx$$

converges, whereas

$$\int_0^\infty |\sin(x^2)| \, dx$$

diverges.

2006.1.2. Let $\mathbf{a} = (a_1, a_2, \dots, a_n)$ be a unit vector, and let $\mathbf{X} = (x_{ij})$ be the $n \times n$ matrix defined by

$$x_{ij} = \delta_{ij} + a_i a_j,$$

where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Compute $\det(\mathbf{X})$.

2006.1.3. Let $h: \mathbb{R} \rightarrow \mathbb{R}$ satisfy

$$\int_0^\infty h(t) \, dt = 1.$$

Evaluate

$$\iint_{\mathbb{R}^2} h(x^2 + 2xy + 5y^2) \, dA.$$

2006.1.4. Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots.$$

Let $\sum a_k$ be a series obtained by rearranging its terms subject to the rule that the relative order among terms of the same sign is not changed. For example,

$$1 + \frac{1}{3} - \frac{1}{2} - \frac{1}{4} + \cdots$$

is allowed, whereas

$$1 - \frac{1}{4} + \frac{1}{3} - \frac{1}{2} + \cdots$$

is not. Among $a_1, a_2, \dots, a_n$, let p_n be the number of positive terms and q_n the number of negative terms.

(a) Prove that $\sum a_k$ converges if and only if the limit

$$\lim_{n \rightarrow \infty} \frac{p_n}{q_n}$$

exists.

(b) If this limit is α , prove that

$$\sum a_k = \ln 2 + \frac{1}{2} \ln \alpha.$$

2007.2.1. Evaluate

$$\int_0^1 \int_0^x (3 - x - y) \, dy \, dx.$$

2007.2.2. Let $C(t) = (\cos t, \sin t)$ for $0 \leq t \leq 2\pi$. Evaluate the line integral

$$\int_C \frac{y \, dx - x \, dy}{x^2 + y^2}.$$

2007.2.3. Find the area of the quadrilateral whose vertices are

$$(0, 0, 0), \quad (1, 2, 3), \quad (3, 1, 2), \quad (7, 4, 7).$$

2007.2.4. Let

$$f(x) = \frac{e^x}{x}, \quad 1 \leq x \leq 2,$$

and let g be the inverse function of f . Evaluate

$$\int_e^{e^2/2} [g(x)]^2 \, dx.$$

2007.2.5. Find the functions $x(t)$ and $y(t)$ satisfying

$$x'(t) = -y(t), \quad y'(t) = x(t), \quad (x(0), y(0)) = (1, 1).$$

2007.2.6. Find the relation among a, b, c that is necessary and sufficient for the system

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= a, \\ 2x_1 + 4x_2 + 6x_3 - 2x_4 &= b, \\ x_1 + x_3 + x_4 &= c \end{aligned}$$

to have a solution.

2007.2.7. Write the matrix

$$\begin{pmatrix} 1 & -3 & 0 \\ 0 & 1 & 3 \\ 2 & -10 & 2 \end{pmatrix}$$

as $\mathbf{LU}$, where $\mathbf{L}$ is lower triangular and $\mathbf{U}$ is upper triangular.

2007.2.8. Three unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ determine a parallelepiped of volume $\frac{1}{2}$. Let θ_{ij} be the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$, and let $\mathbf{A}$ be the 3×3 matrix whose (i, j) -entry is $\cos \theta_{ij}$. Compute $\det(\mathbf{A})$.

2007.2.9. Find the volume of the region in $\mathbb{R}^3$ described by

$$x = (p - q) \cos t, \quad y = (p - q) \sin t, \quad z = p + q,$$

where

$$0 \leq t \leq 2\pi, \quad 1 - \varepsilon \leq p \leq 1 + \varepsilon, \quad -\delta \leq q \leq \delta, \quad \delta < 1 - \varepsilon.$$

2007.2.10. Find the constants a, b that minimize

$$\int_0^\pi [\sin x - (ax + b)]^2 dx.$$

2007.2.11. Let $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ be linearly independent vectors satisfying

$$\|\mathbf{v}_i\| = 1, \quad \mathbf{v}_i \cdot \mathbf{v}_j = -\frac{1}{3} \quad (i \neq j).$$

For vectors $\mathbf{x} \in \mathbb{R}^3$ with $\|\mathbf{x}\| = 1$, find the maximum and minimum values of

$$\sum_{i=1}^3 \|\mathbf{x} - \mathbf{v}_i\|^2.$$

2007.2.12. For real y , evaluate

$$\int_{-\infty}^{\infty} \frac{e^{-2\pi i y x}}{\cosh(\pi x)} dx.$$

2007.2.13. Let $p > 0$ be real. Evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n (1^p + 2^p + 3^p + \cdots + k^p)^{1/(p+1)}.$$

2007.1.1. Find the area of the quadrilateral whose vertices are

$$(0, 0, 0), \quad (1, 2, 3), \quad (3, 1, 2), \quad (7, 4, 7).$$

2007.1.2. Let

$$f(x) = \frac{e^x}{x}, \quad 1 \leq x \leq 2,$$

and let g be the inverse function of f . Evaluate

$$\int_e^{e^2/2} [g(x)]^2 dx.$$

2007.1.3. Let $A = \{1, 2, 3, \dots, n\}$. Suppose k distinct subsets of A , each containing exactly γ elements, are selected. Prove that if every element of A is to belong to at least p of the selected subsets, then

$$k \geq \frac{np}{\gamma}.$$

2007.1.4. Three unit vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^3$ determine a parallelepiped of volume $\frac{1}{2}$. Let θ_{ij} be the angle between $\mathbf{v}_i$ and $\mathbf{v}_j$, and let $\mathbf{A}$ be the 3×3 matrix whose (i, j) -entry is $\cos \theta_{ij}$. Compute $\det(\mathbf{A})$.

2007.1.5. Let $M_{2 \times 2}$ be the vector space of 2×2 matrices, and let $\mathbf{T} \in M_{2 \times 2}$ be invertible. Compute the determinant of the linear map

$$\Phi: M_{2 \times 2} \rightarrow M_{2 \times 2}, \quad \Phi(\mathbf{A}) = \mathbf{TAT}^{-1}.$$

2007.1.6. In triangle ABC , let P be a point on the side BC . Put

$$BP = x, \quad BC = l,$$

let the altitude of the triangle be h , and let

$$\angle APC = \theta(x).$$

Prove that

$$\angle A = \frac{1}{h} \int_0^l \sin^2 \theta(x) \, dx.$$

2007.1.7. Let $\mathbf{A}$ and $\mathbf{B}$ be real matrices. Suppose there exists a complex matrix $\mathbf{S}$ such that

$$\mathbf{A} = \mathbf{S}^{-1} \mathbf{B} \mathbf{S}.$$

Prove that there exists a real matrix $\mathbf{R}$ such that

$$\mathbf{A} = \mathbf{R}^{-1} \mathbf{B} \mathbf{R}.$$

2007.1.8. Let $a_1 > 0$ and

$$a_{n+1} = \ln(1 + a_n).$$

Prove that

$$\lim_{n \rightarrow \infty} n a_n = 2.$$

2007.1.9. Let (a_n) be a sequence such that

$$\sum_{n=1}^{\infty} a_n = 1.$$

Prove that

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n a_k \left(1 - \frac{k^2}{n^2} \right) = 1.$$

2007.1.10. Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ satisfy the following condition: for every triangle Δ of area 1, $F(\Delta)$ is also a triangle of area 1. Prove that for every straight line l , $F(l)$ is also a straight line.

2008.2.1. Find the volume of the tetrahedron whose vertices are

$$(0, 1, 0), \quad (1, 2, 1), \quad (1, 3, 3), \quad (3, 1, 2).$$

2008.2.2. Evaluate

$$\int_0^{\infty} \frac{e^{-x} - e^{-2x}}{x} \, dx.$$

2008.2.3. Evaluate

$$\int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} \int_0^{\sqrt{\pi/2}} xyz \cos(x^2 + y^2 + z^2) \, dx \, dy \, dz.$$

2008.2.4. Let $\alpha > 1$. Prove that the improper integral

$$\int_1^\infty \cos(x^\alpha) \, dx$$

converges.

2008.2.5. Find one matrix $\mathbf{A}$ satisfying

$$\mathbf{A}^2 = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

2008.2.6. Find a function $f(x, y, z)$ satisfying

$$\nabla f(x, y, z) = \left(ye^{xy+yz}, (x+z)e^{xy+yz}, ye^{xy+yz} \right).$$

2008.2.7. Find the maximum value of

$$f(x, y) = 3x^2 + 4xy + y^2$$

on the set

$$\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

2008.2.8. Find all 3×1 column vectors $\mathbf{u}$ and $\mathbf{v}$ satisfying

$$\begin{pmatrix} 5 & 2 & 3 \\ 15 & 6 & 9 \\ 10 & 4 & 6 \end{pmatrix} = \mathbf{u}\mathbf{v}^\top.$$

2008.2.9. Let $\mathbf{A} = (a_{ij})$ be an $n \times n$ matrix satisfying

$$\sum_{i=1}^n a_{ij} = 0$$

for every $j = 1, 2, \dots, n$. Prove that $\mathbf{A}$ is not invertible.

2008.2.10. Let (ϕ_n) be a sequence of functions defined on $[0, 1]$ satisfying

(a) $\phi_0(t) = 0$;

(b)

$$\phi_n(t) = 1 + 2 \sin \frac{t}{2} - \frac{1}{2} \int_0^t \phi_{n-1}(s) \, ds, \quad n = 1, 2, \dots$$

Suppose that $\phi_n \rightarrow \phi$ on $[0, 1]$ and that ϕ is differentiable. Find $\phi(t)$.

2008.2.11. Find all polynomial functions $u(x, y)$ satisfying the following conditions.

(a) $u_{xx} + u_{yy} = 0$.

(b) For every $x \in \mathbb{R}$,

$$u(x, 0) = u(x, \sqrt{3}x) = 0.$$

(c) For every $x > 0$ and $0 < y < \sqrt{3}x$,

$$u(x, y) > 0.$$

(d) There exists a positive integer n such that, for every $t \in \mathbb{R}$,

$$u(tx, ty) = t^n u(x, y).$$

2008.2.12. For every $n = 0, 1, 2, 3, \dots$, let a_n be either 0 or 1. Let P and Q be polynomials with $\deg P < \deg Q$, and suppose that for $|x| < 1$,

$$\sum_{n=0}^{\infty} a_n x^n = \frac{P(x)}{Q(x)}.$$

Prove that there exists a positive integer p such that

$$a_{n+p} = a_n$$

for every $n \geq 0$.

2008.2.13. For the four data points

$$(0, 0), \quad (1, 1), \quad (3, 2), \quad (4, 5),$$

find the line $y = ax + b$ that minimizes the sum of squared errors

$$E^2 = \sum_{i=1}^4 (y_i - ax_i - b)^2.$$

2008.1.1. Find the volume of the tetrahedron whose vertices are

$$(0, 1, 0), \quad (1, 2, 1), \quad (1, 3, 3), \quad (3, 2, 1).$$

2008.1.2. Let $0 < a, b, c, d < 1$. Prove the inequality

$$\begin{aligned} \sqrt{(1-ab)(1-cd)} &\geq \sqrt{ac(1-b)(1-d)} + \sqrt{bd(1-a)(1-c)} \\ &\quad + \sqrt{(1-a)(1-b)(1-c)(1-d)}. \end{aligned}$$

2008.1.3. Evaluate

$$\int_0^{\infty} \frac{e^{-x} - e^{-2x}}{x} dx.$$

2008.1.4. Let $\alpha > 1$. Prove that the improper integral

$$\int_1^{\infty} \cos(x^\alpha) dx$$

converges.

2008.1.5. Evaluate

$$\sin \frac{\pi}{10} \sin \frac{2\pi}{10} \cdots \sin \frac{9\pi}{10}.$$

2008.1.6. Let $\mathbf{A}$ be a 3×3 matrix all of whose entries are positive, and suppose that $\mathbf{A}^{-1}$ exists. If exactly six entries of $\mathbf{A}^{-1}$ are positive, prove that exactly three entries of $\mathbf{A}^{-1}$ are negative.

2008.1.7. Let (ϕ_n) be a sequence of functions defined on $[0, 1]$ satisfying

- (a) $\phi_0(t) = 0$;
- (b)

$$\phi_n(t) = 1 + 2 \sin \frac{t}{2} - \frac{1}{2} \int_0^t \phi_{n-1}(s) ds, \quad n = 1, 2, \dots$$

Show that (ϕ_n) converges on $[0, 1]$, and determine

$$\phi = \lim_{n \rightarrow \infty} \phi_n.$$

2008.1.8. For every $n = 0, 1, 2, 3, \dots$, let a_n be either 0 or 1. Let P and Q be polynomials with $\deg P < \deg Q$, and suppose that for $|x| < 1$,

$$\sum_{n=0}^{\infty} a_n x^n = \frac{P(x)}{Q(x)}.$$

Prove that there exists a positive integer p such that

$$a_{n+p} = a_n$$

for every $n \geq 0$.

2008.1.9. Let $p_i, q_i > 0$ and suppose that

$$\sum_{i=1}^n p_i = \sum_{i=1}^n q_i = 1.$$

Prove that

$$\frac{1}{2} \sum_{i=1}^n (p_i - q_i)^2 \leq \sum_{i=1}^n p_i \ln \left(\frac{p_i}{q_i} \right).$$

2008.1.10. Let (x_n) be a sequence satisfying

$$x_m > 0, \quad x_{m+n} \leq x_m + x_n, \quad m, n = 1, 2, 3, \dots$$

Prove that the sequence (x_n/n) converges.